running stuff on rc

a) using `sinteractive` on our partition

```
# start a job with 20 cpu cores, 4g ram-per-cpu core, and one "full" h100 gpu:
sinteractive --partition=blanca-blast-lecs --account=blanca-blast-lecs --qos=blanca-blast-lecs --nodes=1 --ntasks=20 --mem-per-cpu=4000m --gres=gpu:h100_7g.80gb
```

maria said that we should use 5 `ntasks` per 20 gb of gpu usage. therefore, the complete one (80 gb) should have 20 workers. however, rohan might mention u in the slack channel if u do this lol.

alternative (in case that the 80 gb is being used by someone else) - using `sinteractive` on the `clearlab` partition

both partitions work for the transcription step. the `clearlab` one is not useful for the diarization pipeline since it needs at least 80 gb of memory, which none of those gpus have. what i did, for speeding things up, was using all of these gpus - plus the ones in our partition.

```
sinteractive --partition=blanca-clearlab1 --qos=blanca-clearlab1 --account=blanca-clearlab1 --nodelist=bgpu-g4-u30 --gpu=1
```

b) add the gpu to the environment

```
export CUDA_VISIBLE_DEVICES="0"
```

c) load the modules needed

```
module load cuda/12.1.1.lua
```

```
module load ffmpeg/4.4
```

```
module load uv
```

d) add the dependencies needed for correct gpu detection/usage

```
uv sync --extra gpu
```

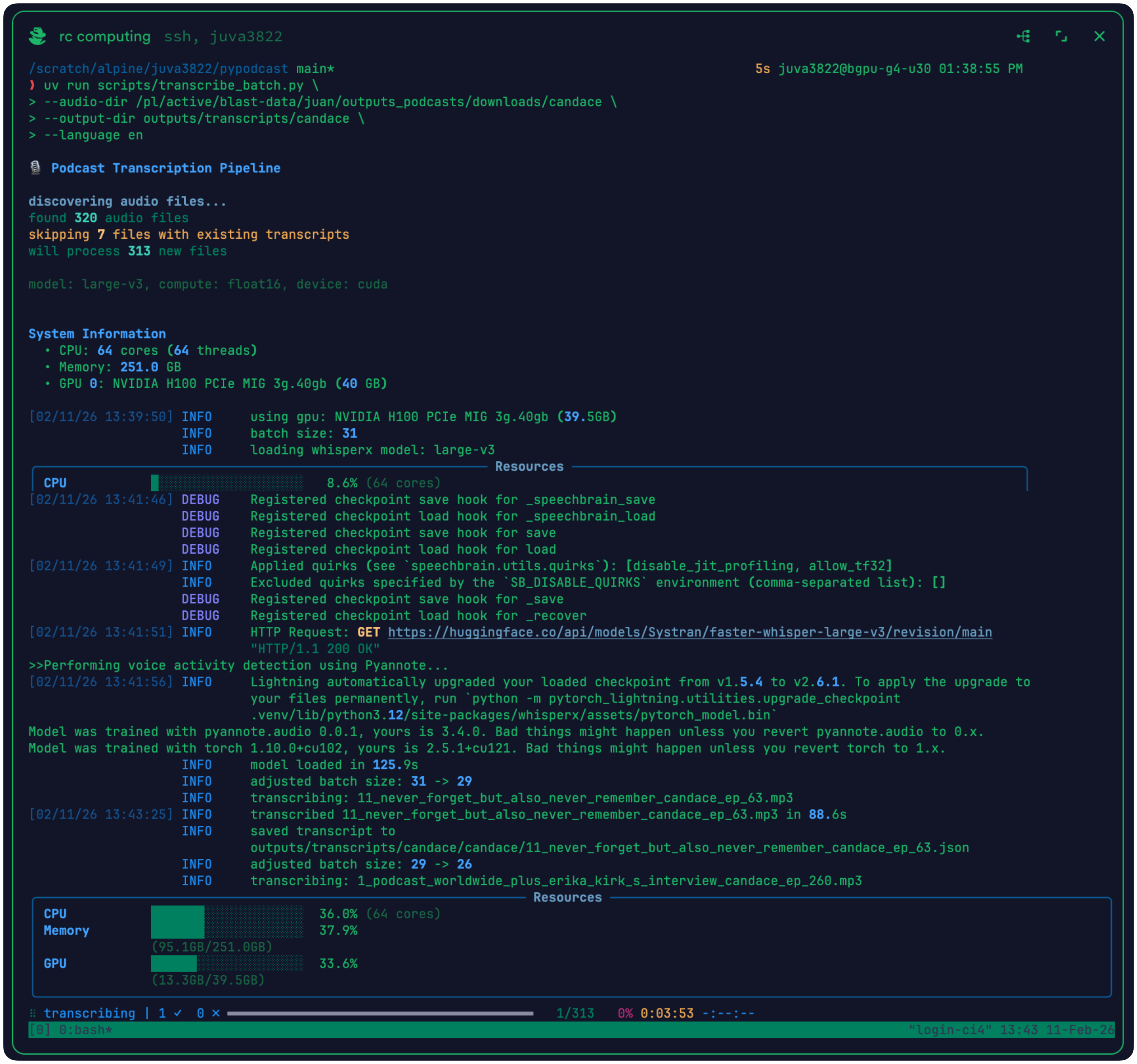
e) transcription

```
wget -O /scratch/alpine/juva3822/.torchcache/hub/checkpoints/wav2vec2_fairseq_base_ls960_asr_ls960.pth https://download.pytorch.org/torchaudio/models/wav2vec2_fairseq_base_ls960_asr_ls960.pth
```

the path after `--audio-dir` is where the audio files i downloaded are stored. u can use those for testing, or use the ones u wanna process (ofc, lol).

```
uv run scripts/transcribe_batch.py \
 --audio-dir /pl/active/blast-data/juan/outputs_podcasts/downloads/candace \
 --output-dir outputs/transcripts/candace \
 --language en
```

this is what i'm seeing after running the previous command:



```
rc computing  ssh, juva3822
/scratch/alpine/juva3822/pypodcast main*                                    5s juva3822@bgpu-g4-u30 01:38:55 PM
❯ uv run scripts/transcribe_batch.py \
> --audio-dir /pl/active/blast-data/juan/outputs_podcasts/downloads/candace \
> --output-dir outputs/transcripts/candace \
> --language en

🎙 Podcast Transcription Pipeline

discovering audio files...
found 320 audio files
skipping 7 files with existing transcripts
will process 313 new files

model: large-v3, compute: float16, device: cuda


System Information
  • CPU: 64 cores (64 threads)
  • Memory: 251.0 GB
  • GPU 0: NVIDIA H100 PCIe MIG 3g.40gb (40 GB)

[02/11/26 13:39:50] INFO     using gpu: NVIDIA H100 PCIe MIG 3g.40gb (39.5GB)
                    INFO     batch size: 31
                    INFO     loading whisperx model: large-v3
──────────────────────────────── Resources ────────────────────────────────
 CPU                           8.6% (64 cores)
[02/11/26 13:41:46] DEBUG    Registered checkpoint save hook for _speechbrain_save
                    DEBUG    Registered checkpoint load hook for _speechbrain_load
                    DEBUG    Registered checkpoint save hook for save
                    DEBUG    Registered checkpoint load hook for load
[02/11/26 13:41:49] INFO     Applied quirks (see `speechbrain.utils.quirks`): [disable_jit_profiling, allow_tf32]
                    INFO     Excluded quirks specified by the `SB_DISABLE_QUIRKS` environment (comma-separated list): []
                    DEBUG    Registered checkpoint save hook for _save
                    DEBUG    Registered checkpoint load hook for _recover
[02/11/26 13:41:51] INFO     HTTP Request: GET https://huggingface.co/api/models/Systran/faster-whisper-large-v3/revision/main
                             "HTTP/1.1 200 OK"
>>Performing voice activity detection using Pyannote...
[02/11/26 13:41:56] INFO     Lightning automatically upgraded your loaded checkpoint from v1.5.4 to v2.6.1. To apply the upgrade to
                             your files permanently, run `python -m pytorch_lightning.utilities.upgrade_checkpoint
                             .venv/lib/python3.12/site-packages/whisperx/assets/pytorch_model.bin`
Model was trained with pyannote.audio 0.0.1, yours is 3.4.0. Bad things might happen unless you revert pyannote.audio to 0.x.
Model was trained with torch 1.10.0+cu102, yours is 2.5.1+cu121. Bad things might happen unless you revert torch to 1.x.
                    INFO     model loaded in 125.9s
                    INFO     adjusted batch size: 31 -> 29
                    INFO     transcribing: 11_never_forget_but_also_never_remember_candace_ep_63.mp3
[02/11/26 13:43:25] INFO     transcribed 11_never_forget_but_also_never_remember_candace_ep_63.mp3 in 88.6s
                    INFO     saved transcript to
                             outputs/transcripts/candace/candace/11_never_forget_but_also_never_remember_candace_ep_63.json
                    INFO     adjusted batch size: 29 -> 26
                    INFO     transcribing: 1_podcast_worldwide_plus_erika_kirk_s_interview_candace_ep_260.mp3
──────────────────────────────── Resources ────────────────────────────────
 CPU                          36.0% (64 cores)
 Memory                       37.9%
                  (95.1GB/251.0GB)
 GPU                          33.6%
                  (13.3GB/39.5GB)

⠿ transcribing | 1 ✓  0 ✕ ━━━━━━━━━━━━━━━━━━━━  1/313  0% 0:03:53 -:--:--
[0] 0:bash*                                                              "login-ci4" 13:43 11-Feb-26
```

> ⓘ **note**
>
> if the gpu utilization shows 0.0% or very low usage, that means that $\texttt{pynvml}$ is likely not returning utilization for mig instances.
>
> mig partitions often report 0% via `nvmldevicegetutilizationrates` because $\texttt{nvml}$ doesn't support per-mig-instance utilization tracking. this is a known nvidia limitation 😒 .

g) diarization

this is a sample command i used for my pipeline. i leave my hf token here, so u can use it if u want to. u should adapt the number of `workers` depending on how many cpus u asked for when running `sinteractive … --ntasks=5`.

the `--compile-mode` is just an optimization for running things on (in ?) a gpu.

```
./scripts/diarize_all_podcasts.sh \
    --transcripts-root outputs/transcripts/candace \
    --audio-base-dir  /pl/active/blast-data/juan/outputs_podcasts/downloads \
    --output-dir outputs/diarizations/candace \
    --hf-token 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 \
    --workers 5 \
    --compile-mode max-autotune
```